Retained adjoint-fermion binding with explicit $\hbar$: recoil, quantum corrections and the whole-state current

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Abstract

We study two fermionic endpoints bound to a retained adjoint fermion by regulated contact interactions, with no direct quark–antiquark force. In a fixed-number, global-color model, dimer binding suffices to prove a trimer below breakup. With physical units and explicit $\hbar$, we distinguish recoil perturbation theory, the semiclassical expansion and contact iterations. The leading recoil energy correction is second order and attractive. For a Gaussian contact in the deep-well regime, we derive the energy through order $\hbar^2$ with exact mass-ratio dependence at each displayed order, together with the correlated harmonic charge radius. Contact binding does not by itself make this current pointlike. These results develop the user-selected retained-carrier model; a relativistic or common string realization remains a separate construction task.

Research scope and provenance. The broader target remains bosonic strings with $q/\bar{q}$ or q/q endpoints for mesons or diquarks, and fermionic strings with $q/\bar{q}$ or $\bar{q}/\bar{q}$ endpoints for leptons or quarks. The main phenomenological question is approximately pointlike scattering of the whole fermionic state. The user now selects the retained adjoint contact model as the most interesting concrete object and requests explicit $\hbar$ in all calculations.

This focused revision develops that selection. It does not identify a contact trimer with a string by definition. Version 0.2, including the warped-current comparison and the scalar-completion study, remains preserved at repository commit `6b73d29`; version 0.1 is preserved at `333e537`. The uploaded originals remain unchanged. Earlier natural-unit calculations are historical comparisons, not the unit convention of this paper. The new calculation uses standard spectral perturbation theory and oscillator methods; no claim of literature priority follows from their application here.

1 A retained adjoint binds both endpoints

We take distinguishable nonrelativistic endpoint species $q, \bar{q}$ with physical masses m, m, and a retained adjoint fermion λ with physical mass M. Constituent numbers are conserved and color is global $SU(3)$. These assumptions define the model, not a relativistic QCD completion. The carrier is present in the bound state; it is not a virtual fermion exchanged between unchanged endpoints.

Throughout, $\hbar$ is explicit, c_{light} denotes the speed of light, positions have units of length and momenta have units of momentum:

$$\mathbf{p} = -i\hbar\nabla, \qquad i\hbar\partial_t\Psi = H\Psi, \qquad [G] = \text{energy}\,\text{length}^3. \tag{1}$$

The color matrices are dimensionless and obey $\mathrm{tr}(T^aT^b) = \delta^{ab}/2$. The unique singlet in $\mathbf{3}\otimes\bar{\mathbf{3}}\otimes\mathbf{8}$ has

$$T_q \cdot T_{\bar{q}} = \frac{1}{6}, \qquad T_q \cdot T_\lambda = T_{\bar{q}} \cdot T_\lambda = -\frac{3}{2}. \tag{2}$$

There is no direct $q\bar{q}$ potential. With $\mathbf{x} = \mathbf{r}_q - \mathbf{r}_\lambda$ and $\mathbf{y} = \mathbf{r}_{\bar{q}} - \mathbf{r}_\lambda$, set

$$f_R(r) = \frac{e^{-r^2/R^2}}{\pi^{3/2}R^3}, \qquad V = -\frac{3G}{2}[f_R(x) + f_R(y)], \qquad G > 0, \tag{3}$$

$$\mu = \frac{mM}{m+M}, \qquad H_{\rm rel} = \frac{p_x^2 + p_y^2}{2\mu} + \frac{\mathbf{p}_x \cdot \mathbf{p}_y}{M} + V. \tag{4}$$

Equivalently the kinetic operator is $-\hbar^2(\nabla_x^2 + \nabla_y^2)/(2\mu) - \hbar^2 \nabla_x \cdot \nabla_y / M$. The two contacts are number-conserving four-fermion density interactions. No propagating bosonic binding field is included in this Hamiltonian. Their ultraviolet origin remains unspecified. $R > 0$ is the specified finite regulator length; we do not silently take an unrenormalized three-dimensional delta limit.

The spin-independent model allows endpoint spin zero and total spin 1/2 supplied by λ. Physical spins satisfy $s^2 = 3\hbar^2/4$. Spin-dependent interactions, allowed annihilation channels and gauged color would change the problem and must be supplied explicitly.

1.1 Binding and the correct breakup threshold

For $h_\mu = p^2/(2\mu) - 3Gf_R/2$, a normalized Gaussian orbital of width a gives

$$E_2^{\rm trial}(a) = \frac{3\hbar^2}{4\mu a^2} - \frac{3G}{2\pi^{3/2}(R^2 + a^2)^{3/2}}. \tag{5}$$

At $a^2 = 2R^2$ this proves the sufficient condition

$$\boxed{G > \frac{3\sqrt{3}\,\pi^{3/2}}{4}\frac{\hbar^2 R}{\mu} \quad \Longrightarrow \quad \epsilon_2 < 0.} \tag{6}$$

Here ϵ_2 is the exact dimer ground energy, not its variational estimate. If u is its real normalized orbital, the product $u(x)u(y)$ has $\langle \mathbf{p}_x \cdot \mathbf{p}_y \rangle = 0$ and energy $2\epsilon_2$. Both dimer-plus-spectator thresholds are ϵ_2, so

$$\boxed{E_3 \le 2\epsilon_2 < \epsilon_2 < 0.} \tag{7}$$

The bounded short-range potentials and variational principle establish a trimer below breakup. Removing λ removes both interactions. This is a binding result with no pre-existing $q\bar{q}$ bound state.

Nonrelativistic control requires characteristic momenta much smaller than $mc_{\rm light}$ and $Mc_{\rm light}$, and binding energies small compared with the corresponding rest energies. A useful range choice is $R \gg \hbar/(mc_{\rm light}), \hbar/(Mc_{\rm light})$. The model does not yet generate a light relativistic lepton, a confining flux tube, or protection against annihilation in a larger theory.

2 Three different expansions

Define a depth, a dimensionless contact strength, and a recoil parameter:

$$V_0 = \frac{3G}{2\pi^{3/2}R^3}, \qquad \gamma = \frac{2\mu V_0 R^2}{\hbar^2} = \frac{3\mu G}{\pi^{3/2}\hbar^2 R}, \qquad \eta = \frac{\mu}{M} = \frac{m}{m+M}. \tag{8}$$

The energy unit $E_R = \hbar^2/(2\mu R^2)$ and coordinates $\mathbf{X} = \mathbf{x}/R$, $\mathbf{Y} = \mathbf{y}/R$ give

$$\frac{H_{\rm rel}}{E_R} = -\nabla_X^2 - \nabla_Y^2 - 2\eta \nabla_X \cdot \nabla_Y - \gamma(e^{-X^2} + e^{-Y^2}). \tag{9}$$

This rescaling displays $\hbar$ in E_R and γ; it does not set it to one. Equation (6) becomes $\gamma > 9\sqrt{3}/4$.

Three expansions must be kept distinct: a recoil series in η about two bound dimers; a deep-well semiclassical series in $\varepsilon = \gamma^{-1/2} \propto \hbar$ at fixed physical potential and masses; and a Born series in the contact interaction about free propagation. They have different domains. In particular, the existence of a shallow bound-state pole generally requires resumming the Born series.

3 Recoil perturbation theory about the bound product

At fixed μ, G, R write

$$H(\eta) = H_0 + \eta W, \qquad H_0 = h_\mu(x) + h_\mu(y), \qquad W = \frac{\mathbf{p}_x \cdot \mathbf{p}_y}{\mu}. \tag{10}$$

The unperturbed orbital ground state is $|0\rangle = |u, u\rangle$, with $E_0 = 2\epsilon_2$. Spin degeneracy is inert in this spin-independent Hamiltonian. Reversing $\mathbf{x}$ alone maps $H(\eta)$ unitarily to $H(-\eta)$. Thus the isolated orbital ground-energy branch is even near $\eta = 0$:

$$E_3(\eta) = 2\epsilon_2 - \eta^2 \mathcal{C}_2 + O(\eta^4), \qquad \mathcal{C}_2 = \frac{1}{\mu^2} \sum_{a,b} \frac{|\langle a, b|\mathbf{p}_x \cdot \mathbf{p}_y|u, u\rangle|^2}{\Delta_a + \Delta_b} > 0. \tag{11}$$

Here $\Delta_a = \epsilon_a - \epsilon_2$; the spectral sum includes integrals over continuum states. Each momentum takes the dimer S wave into an $l = 1$ state, and their scalar contraction selects total orbital angular momentum zero. The first-order shift vanishes, while the second-order shift is attractive. The intermediate states need not be bound P waves. The coefficient has units of energy, and its momenta contain the explicit $\hbar$ above.

This is a correction from the motion of the same retained carrier. It does not insert a direct pair potential between the endpoints. The product trial state already proves binding; the correlated wavefunction lowers its energy further. Recoil correlations are an actual dynamical effect even though their first-order energy expectation vanishes.

For subsequent orders, let $\mathcal{Q} = 1 - |0\rangle\langle 0|$ and $\mathcal{R}_0 = \mathcal{Q}(E_0 - H_0)^{-1}\mathcal{Q}$. With intermediate normalization and $|\psi\rangle = \sum_{n\geq 0} \eta^n |\psi_n\rangle$, one has

$$E_n = \langle 0|W|\psi_{n-1}\rangle, \tag{12}$$

$$|\psi_n\rangle = \mathcal{R}_0 \left[W|\psi_{n-1}\rangle - \sum_{j=1}^{n-1} E_j |\psi_{n-j}\rangle \right], \qquad n \geq 1. \tag{13}$$

Odd energy coefficients vanish. This is an all-order prescription, not an evaluation of every coefficient or a claim of convergence up to $\eta = 1$. A small dimer gap can limit its useful domain.

The fixed-μ qualification matters: changing M at fixed endpoint mass m also changes μ. The dimer contribution $2\epsilon_2(\mu)$ then changes, and the total physical heavy-mass expansion need not contain only even powers of $1/M$.

4 An analytic deep-well calculation through order $\hbar^2$

For the specified Gaussian contact take $\gamma \gg 1$. Define

$$\omega^2 = \frac{2V_0}{\mu R^2}, \qquad \varepsilon = \frac{\hbar}{\mu\omega R^2} = \gamma^{-1/2} \ll 1, \qquad S_\eta = \sqrt{1+\eta} + \sqrt{1-\eta}. \tag{14}$$

Hold $0 \le \eta < 1$ fixed in this expansion. Deep binding relative to E_R can coexist with nonrelativistic control if V_0 remains small compared with the constituent rest energies and R is sufficiently large.

In $\mathbf{u}_\pm = (\mathbf{x} \pm \mathbf{y})/\sqrt{2}$, the harmonic Hamiltonian is

$$H_{\rm harm} = -2V_0 + \sum_{\sigma=\pm} \left[\frac{(1+\sigma\eta)p_\sigma^2}{2\mu} + \frac{\mu\omega^2 u_\sigma^2}{2} \right]. \tag{15}$$

Its frequencies are $\omega\sqrt{1\pm\eta}$, and hence

$$E_{3,\rm harm} = -2V_0 + \frac{3\hbar\omega}{2} S_\eta. \tag{16}$$

This resums recoil exactly within the harmonic approximation. Relative to twice the harmonic dimer energy,

$$E_{3,\rm harm} - 2\epsilon_{2,\rm harm} = -\frac{3\hbar\omega}{8}\eta^2 - \frac{15\hbar\omega}{128}\eta^4 + O(\hbar\omega\eta^6). \tag{17}$$

The first anharmonic term is $V_4 = -V_0(x^4+y^4)/(2R^4)$, where $x^4 = (\mathbf{x}^2)^2$. The variance of each Cartesian component of $\mathbf{x}$ or $\mathbf{y}$ in the harmonic ground state is $s = \hbar S_\eta/(4\mu\omega)$. Since $\langle x^4\rangle = 15s^2$, first-order perturbation theory in V_4 gives

$$\boxed{E_3 = -2V_0 + \frac{3\hbar\omega}{2} S_\eta - \frac{15\hbar^2}{32\mu R^2} S_\eta^2 + O(V_0\varepsilon^3).} \tag{18}$$

Thus the classical well depth, harmonic zero-point energy and first anharmonic correction are explicitly orders $\hbar^0, \hbar^1, \hbar^2$ at fixed V_0, R, μ, η. Sextic first-order perturbation and quartic second-order perturbation both first enter at order $\hbar^3$; retaining only one would be incomplete.

The corresponding dimer expansion is

$$\epsilon_2 = -V_0 + \frac{3\hbar\omega}{2} - \frac{15\hbar^2}{16\mu R^2} + O(V_0\varepsilon^3). \tag{19}$$

In particular, the coefficient of η^2 in the combined expansion is

$$E_3 - 2\epsilon_2 = \left[-\frac{3\hbar\omega}{8} + \frac{15\hbar^2}{32\mu R^2} + O(\hbar\omega\varepsilon^2) \right] \eta^2 + O(\hbar\omega\eta^4). \tag{20}$$

The attractive curvature agrees with Eq. (11) in the controlled regime $\varepsilon \ll 1$. These formulas concern the chosen finite Gaussian regulator; they are not universal coefficients of every short-range force.

5 The whole-state current with physical momentum transfer

Take an electrically neutral λ and endpoint charges e_1, e_2 with $e_1 + e_2 \neq 0$. Equal endpoint masses and symmetric dynamics make their one-body elastic form factors equal, even when their charges differ. A charged state may require different endpoint flavors; the binding proof did not require a nonzero electric charge.

The leading constituent current measures

$$\mathbf{r}_q - \mathbf{R}_{\rm cm} = \frac{(m+M)\mathbf{x} - m\mathbf{y}}{2m+M} = \frac{\mathbf{x} - \eta\mathbf{y}}{1+\eta}, \qquad F(\mathbf{Q}) = \left\langle e^{i\mathbf{Q}\cdot(\mathbf{r}_q - \mathbf{R}_{\rm cm})/\hbar} \right\rangle. \tag{21}$$

Here $\mathbf{Q}$ is physical momentum transfer, not wave number. With $r_E^2 = \langle |\mathbf{r}_q - \mathbf{R}_{\rm cm}|^2 \rangle$,

$$F(Q) = 1 - \frac{Q^2 r_E^2}{6\hbar^2} + O\left(\frac{Q^4 \langle r^4 \rangle}{\hbar^4}\right). \tag{22}$$

For a neutral total state the normalized electric form factor is unavailable; other currents or multipoles must be specified.

The harmonic correlated state gives

$$\langle x^2 \rangle_{\rm harm} = \frac{3\hbar}{4\mu\omega} S_\eta, \qquad \langle \mathbf{x} \cdot \mathbf{y} \rangle_{\rm harm} = \frac{3\hbar}{4\mu\omega} D_\eta, \tag{23}$$

$$D_\eta = \sqrt{1+\eta} - \sqrt{1-\eta}, \qquad r_{E,\rm harm}^2 = \frac{3\hbar[(1+\eta^2)S_\eta - 2\eta D_\eta]}{4\mu\omega(1+\eta)^2}. \tag{24}$$

The positive cross-correlation is generated by carrier recoil. It reduces the mean-square radius relative to a distribution with the same marginal variance but zero cross-correlation. The total change also includes the change in that marginal variance and in the center-of-mass weights.

The Gaussian harmonic result is

$$F_{\rm harm}(Q) = \exp\left[-\frac{Q^2 r_{E,\rm harm}^2}{6\hbar^2}\right]. \tag{25}$$

For a single dressed current insertion, after matching the point-particle kinematics and assuming no independent magnetic or transition form factors, the Born rate ratio is $|F|^2$. The illustrative condition $|F|^2 \geq 1 - \delta$ gives

$$r_{E,\rm harm}^2 \leq -\frac{3\hbar^2}{Q^2} \log(1-\delta), \qquad 0 < \delta < 1. \tag{26}$$

Contact binding alone therefore does not remove size from the current. In particular, a mean-square radius proportional to $\hbar$ is insufficient by itself for pointlike scattering at fixed physical Q: the resolution length is $\hbar/Q$. The harmonic approximation to the current is controlled for fixed $Q\sqrt{r_E^2}/\hbar$ in the deep-well expansion, especially near the pointlike window. Its exponentially small large-Q tail is not a uniform prediction of the finite Gaussian well as $\hbar \to 0$.

6 Contact iterations are not the same as the semiclassical series

Choose momentum normalization $\langle \mathbf{r}|\mathbf{p}\rangle = e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}$, with completeness measure $d^3p/(2\pi\hbar)^3$. The dimer potential kernel is

$$V(\mathbf{p}', \mathbf{p}) = -\frac{3G}{2} \exp\left[-\frac{R^2|\mathbf{p}' - \mathbf{p}|^2}{4\hbar^2}\right]. \tag{27}$$

Its scattering operator obeys

$$T(\mathbf{p}', \mathbf{p}; E) = V(\mathbf{p}', \mathbf{p}) + \int \frac{d^3k}{(2\pi\hbar)^3} \frac{V(\mathbf{p}', \mathbf{k})\, T(\mathbf{k}, \mathbf{p}; E)}{E - k^2/(2\mu) + i0}. \tag{28}$$

Every factor has physical units: V and T here have units of energy times volume. Repeated propagation in this equation is organized by the contact strength and the energy denominator, not by assigning an extra positive power of $\hbar$ to every iteration. At fixed physical G, R, μ, $\gamma \propto \hbar^{-2}$; taking $\hbar$ small makes this interaction strong in dimensionless Schrödinger units.

A bound-state pole is a nonperturbative feature of the full resolvent. Quantum-field-theory loop counting instead follows from a specified action in $\exp(iS/\hbar)$ and a stated field/coupling normalization.

The present fixed-number Hamiltonian has no vacuum pair-creation loops. A relativistic completion would have to specify them before their $\hbar$ orders could be calculated. We have not computed such a completion's radiative corrections.

7 Result and next calculation

The retained adjoint contact model binds both endpoints without a direct endpoint force. Keeping $\hbar$ explicit reveals three separate controls and supplies two further results: attractive second-order recoil for the bound product, and the energy through $\hbar^2$ with recoil resummed at each displayed semiclassical order. The same harmonic state has a nonzero, calculable whole-state charge radius.

The next quantitative target is the finite-range trimer current beyond the harmonic approximation, in a stated regime. A useful first extension is the leading anharmonic correction to r_E^2; a different near-threshold target would require resummed two- and three-body propagation. Neither needs an immediate three-generation model. A string or Seiberg-dual completion remains a possible bridge to the broader correspondence, with an explicit operator dictionary and physical currents.

The previous scalar-completion study produces spin-dependent contacts rather than the spin-independent interaction assumed here. For physical spins its projectors are $P_0 = 1/4 - \mathbf{s}_i \cdot \mathbf{s}_j/\hbar^2$ and $P_1 = 1 - P_0$. Its binding proof cannot be imported unchanged. This limitation does not alter the result for the Hamiltonian in Eq. (4).

Reproducibility and review scope

The derivations in this revision are analytic; no new numerical physics script was required. A second research agent reviewed only the new recoil, semiclassical and current formulas. Routine document checks are assigned to a smaller model, following the user's two-level validation policy. Such reviews test consistency, not experimental correctness or novelty.

The living source and PDF are in

```
discussions/2026-10-11-quark-antiquark-open-string/manuscript/
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The research notebook records the model assumptions, source exposure, prior results and next task. The historical motivations include Schwarz's dual quark–gluon model [1], with Seiberg duality [2] retained as a possible later connection. Neither reference proves the contact interaction or its intended string interpretation.

References

[1] J. H. Schwarz, *Dual quark-gluon model of hadrons*, Phys. Lett. B **37**, 315–319 (1971), doi:10.1016/0370-2693(71)90028-1. Metadata and abstract verified previously; full primary text not yet reviewed.

[2] N. Seiberg, *Electric-magnetic duality in supersymmetric non-Abelian gauge theories*, arXiv:hep-th/9411149. Relevant sections were read in the preceding notebook work; no new operator matching is claimed here.